

Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.24)

Metric

$$ds^2 = -\frac{16m^3 e^{\left(\frac{r(u,v)}{2m}\right)}}{r(u,v)} du \otimes dv - \frac{16m^3 e^{\left(\frac{r(u,v)}{2m}\right)}}{r(u,v)} dv \otimes du + r(u,v)^2 d\theta \otimes d\theta + r(u,v)^2 \sin^2(\theta) d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (u, v, \theta, \phi)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(\frac{16m^3 e^{\left(\frac{r(u,v)}{2m}\right)}}{r(u,v)} \right) dv$$

$$m = \left(\frac{1}{2} \sqrt{2} r(u,v) \right) d\theta + \left(\frac{1}{2} i \sqrt{2} r(u,v) \sin(\theta) \right) d\phi$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} r(u,v) \right) d\theta + \left(-\frac{1}{2} i \sqrt{2} r(u,v) \sin(\theta) \right) d\phi$$

Contravariant components

$$l = \left(-\frac{e^{\left(-\frac{r(u,v)}{2m}\right)} r(u,v)}{16m^3} \right) dv$$

$$n = (-1) \, du$$

$$m = \left(\frac{\sqrt{2}}{2 \, r(u, v)} \right) d\theta + \left(\frac{i \, \sqrt{2}}{2 \, r(u, v) \sin(\theta)} \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2 \, r(u, v)} \right) d\theta + \left(-\frac{i \, \sqrt{2}}{2 \, r(u, v) \sin(\theta)} \right) d\phi$$

Directional Derivatives

$$D = -\frac{e^{\left(-\frac{r(u, v)}{2 \, m}\right)} r(u, v)}{16 \, m^3} \partial_v$$

$$\Delta = -\partial_u$$

$$\delta = \frac{\sqrt{2}}{2 \, r(u, v)} \partial_\theta + \frac{i \, \sqrt{2}}{2 \, r(u, v) \sin(\theta)} \partial_\phi$$

$$\bar{\delta} = \frac{\sqrt{2}}{2 \, r(u, v)} \partial_\theta - \frac{i \, \sqrt{2}}{2 \, r(u, v) \sin(\theta)} \partial_\phi$$

Spin Coefficients

$$\rho = \frac{e^{\left(-\frac{r(u, v)}{2 \, m}\right)} \frac{\partial}{\partial v} r(u, v)}{16 \, m^3}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\frac{\partial}{\partial u} r(u, v)}{r(u, v)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2} \cos(\theta)}{4 r(u, v) \sin(\theta)}$$

$$\beta = \frac{\sqrt{2} \cos(\theta)}{4 r(u, v) \sin(\theta)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{(2m - r(u, v)) \frac{\partial}{\partial u} r(u, v)}{4mr(u, v)}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{\left(2m \frac{\partial}{\partial v} r(u, v)^2 - r(u, v) \frac{\partial}{\partial v} r(u, v)^2 + 2mr(u, v) \frac{\partial^2}{(\partial v)^2} r(u, v)\right) e\left(-\frac{r(u, v)}{m}\right)}{512m^7}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\left(32 m^4 e^{\left(\frac{r(u,v)}{2m}\right)} + 2 m r(u,v)^2 \frac{\partial^2}{\partial u \partial v} r(u,v) - r(u,v)^3 \frac{\partial^2}{\partial u \partial v} r(u,v) + 2 m r(u,v) \frac{\partial}{\partial u} r(u,v) \frac{\partial}{\partial v} r(u,v)\right) e^{\left(-\frac{r(u,v)}{2m}\right)}}{128 m^4 r(u,v)^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{2 m \frac{\partial}{\partial u} r(u,v)^2 - r(u,v) \frac{\partial}{\partial u} r(u,v)^2 + 2 m r(u,v) \frac{\partial^2}{(\partial u)^2} r(u,v)}{2 m r(u,v)^2}$$

$$\Lambda = \frac{\left(32 m^4 e^{\left(\frac{r(u,v)}{2m}\right)} + 6 m r(u,v)^2 \frac{\partial^2}{\partial u \partial v} r(u,v) + r(u,v)^3 \frac{\partial^2}{\partial u \partial v} r(u,v) + 6 m r(u,v) \frac{\partial}{\partial u} r(u,v) \frac{\partial}{\partial v} r(u,v)\right) e^{\left(-\frac{r(u,v)}{2m}\right)}}{384 m^4 r(u,v)^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{\left(32 m^4 e^{\left(\frac{r(u,v)}{2m}\right)} - 6 m r(u,v)^2 \frac{\partial^2}{\partial u \partial v} r(u,v) + r(u,v)^3 \frac{\partial^2}{\partial u \partial v} r(u,v) + 6 m r(u,v) \frac{\partial}{\partial u} r(u,v) \frac{\partial}{\partial v} r(u,v)\right) e^{\left(-\frac{r(u,v)}{2m}\right)}}{192 m^4 r(u,v)^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"